

Used Car Sales Analysis

By Zaquel Cossa

Table of Contents

1. Introduction
2. Dataset Overview
3. Data Cleaning Steps
4. Excel Analysis / Pivot Tables
5. DAX
6. PowerBI report
7. Key Insights
8. Conclusion

Introduction

This project explores patterns and pricing behavior in the used car market using a synthetic dataset of 10,000 records from 2015 to 2024.

The goal is to uncover insights on:

- What types of cars sell more often
- How pricing and discounts vary by manufacturer and car type
- What trends exist in age and pricing of sold vehicles

Tools & Workflow:

- **Excel** for initial data cleaning (data types, blanks, formatting)
- **Power BI** for dashboard creation using DAX and visualizations
- **SQL & Excel formulas** for calculated columns and summaries

Dataset Overview

Synthetic dataset of 10,000 used car sales records (2015–2024)

Kaggle : <https://www.kaggle.com/datasets/sandeep1080/used-car-sales>

Key Attributes (25 Columns Total):

- Car Details:
Manufacturer Name, Car Name, Car Type, Manufactured Year, Engine Power (HP), Mileage (KM)
- Transaction Info:
Price-\$ (Listed Price), Sold Price-\$, Purchased Price-\$, Car Sale Status, Sold Date, Purchased Date, Margin-%, Sales Commission-\$
- Customer & Dealer Info: Distributor Name, Location, Sales Agent Name, Sales Rating

Notes: Sale prices of unsold cars are recorded as 0

Raw Data

ID	Distributor Name	Location	Car Name	Manufacturer Name	Car Type	Color	Gearbox	Number of Seats	Number of Doors	Energy
O2KE17	Carmudi	California	Fortuner	Toyota	SUV	Gray	Automatic	8	5	Hybrid
EPMP8	Carousell	Philadelphia	Creta	Hyundai	Hatchback	Blue	Automatic	5	5	Hybrid
SQKXAP	Carsome	North Carolina	Scorpio	Mahindra	SUV	Gray	Automatic	5	5	Hybrid
PWP2QK	Trivett	North Carolina	Plato	Prazo	Convertible	Gray	Automatic	2	2	Petrol
FNDDKM	Zupps	Portland	Dzire	Maruti	Sedan	Red	Automatic	5	5	Electric
I5D584	Zupps	Denver	Fortuner	Toyota	SUV	Red	Automatic	8	5	Electric

Manufactured Year	Price-\$	Mileage-KM	Engine Power-HP	Purchased Date	Car Sale Status	Sold Date	Purchased Price-\$	Sold Price-\$	Margin-%	Sales Agent Name	Sales Rating	Sales Commission-\$	Feedback
2021	8800	67948	200	10/26/2022	Un Sold	1/1/1970	8296	0	0	Pranav	1	0	Average
2018	7600	34637	113	8/25/2017	Sold	3/3/2021	5659	4770	-16	Vihaan	5	0	Good
2016	9300	33274	120	6/13/2018	Un Sold	1/1/1970	8430	0	0	Aarush	4	0	Good
2021	8800	46250	250	5/14/2023	Sold	4/2/2024	6919	7942	15	Anushka	1	205	Average
2019	8500	29323	100	8/24/2022	Un Sold	1/1/1970	6864	0	0	Pavan	3	0	Poor
2021	9300	38716	200	6/5/2022	Sold	11/2/2024	8533	9929	16	Preeti	5	279	Excellent

Data Cleaning Steps

1. Changed data types (e.g., dates, numbers, text) to correct formats
2. Filled or handled missing values where possible
3. Created new columns like “Discount Percentage”
4. Standardized column names for clarity and consistency
5. Filtered out unsold cars (where Sold Price = 0) for certain analyses

Cleaned Data

ID	Distributor Name	Location	Car Name	Manufacturer Name	Car Type	Color
O2KE17	Carmudi	California	Fortuner	Toyota	SUV	Gray
EPMP8	Carousell	Philadelphia	Creta	Hyundai	Hatchback	Blue
SQKXAP	Carsome	North Carolina	Scorpio	Mahindra	SUV	Gray
PWP2OK	Trivett	North Carolina	Plato	Prazo	Convertible	Gray

Gearbox	Number of Seats	Number of Doors	Energy	Car Age	Manufactured Year	Price-\$	Mileage-KM
Automatic	8	5	Hybrid	4	2021	8800	67948
Automatic	5	5	Hybrid	7	2018	7600	34637
Automatic	5	5	Hybrid	9	2016	9300	33274
Automatic	2	2	Petrol	4	2021	8800	46250

Engine Power-HP	Purchased Date	Car Sale Status	Sold Date	Purchased Price-\$	Sold Price-\$
200	10/26/2022	Un Sold	1/1/1970	8296	0
113	8/25/2017	Sold	3/3/2021	5659	4770
120	6/13/2018	Un Sold	1/1/1970	8430	0

Margin-%	Sales Agent Name	Sales Rating	Sales Commission-\$	Feedback
0	Pranav	1	0	Average
-16	Vihaan	5	0	Good
0	Aarush	4	0	Good

Pivot Tables

Avg Margin % by Sales Agent

Total Count of Car Type

Row Labels	Count of Car Type
SUV	3167
Hatchback	3098
Truck	1601
Sedan	1454
Convertible	680
Grand Total	10000

Average Price by Manufacturer

Row Labels	Average of Price-\$
Praza	9630.441176
Mahindra	8576.846058
Renault	8576.232617
Toyota	8358.747579
Tata	8091.686747
Maruti	7423.287671
Hyundai	7113.060686
Kia	7071.927555
Grand Total	7975.1

Row Labels	Average of Margin-%
Rithvik	1.041322314
Pavan	1.012987013
Vishwajeet	0.825065274
Aarush	0.767810026
Shilpa	0.76127321
Preeti	0.742857143
Shailaja	0.707253886
Ram	0.705882353
Swapna	0.638888889
Karan	0.610666667
Sai	0.536363636
Shiva	0.427350427
Piyush	0.402234637
Bhavani	0.344473008
Anushka	0.343243243
Swathi	0.313829787
Supriya	0.308093995
Pranav	0.284507042
Narayan	0.278074866
Arjuna	0.237536657
Shruthi	0.147398844
Krishna	0.117073171
Sweta	0.102564103
Hariom	0.100515464
Aditya	-0.007481297
Vihaan	-0.08908046
Dhruv	-0.116094987
Grand Total	0.4282

DAX

```
AvgPrice = AVERAGE(Table1_1[Price-$])
```

```
Discount % =  
IF(  
    UsedCars[Car Sale Status] = "Sold" && UsedCars[Sold Price-$] <> 0,  
    DIVIDE('UsedCars'[Price-$] - UsedCars[Sold Price-$], UsedCars[Price-$]),  
    BLANK()  
)
```

```
Avg_Discount_Sold = AVERAGE(UsedCars[Discount %])
```

Used Car Sales Analysis

15M

Total Revenue

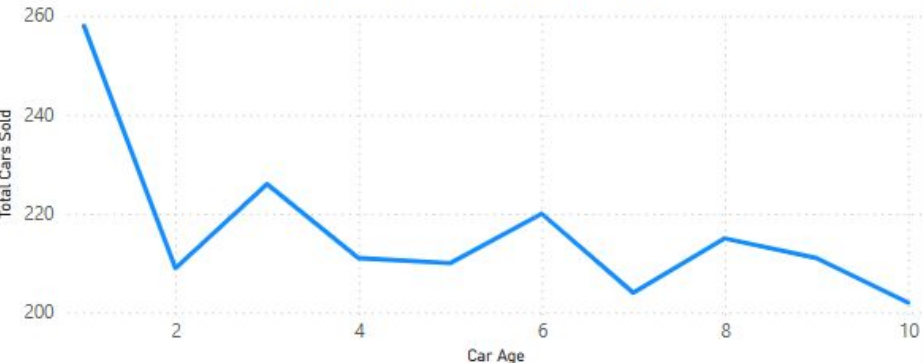
1.49K

Average Sold Price

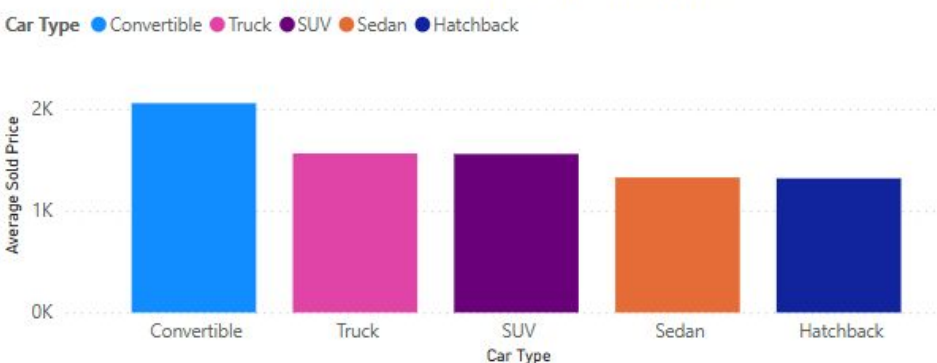
2166

Total Cars Sold

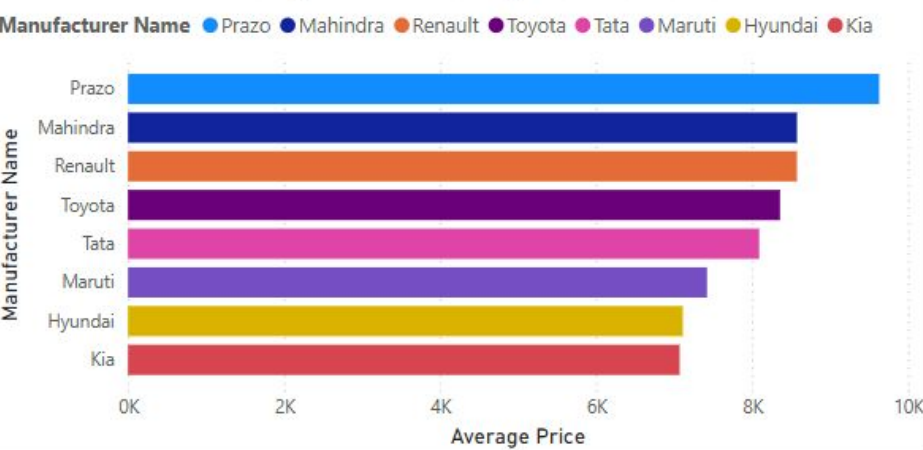
Total Cars Sold by Car Age



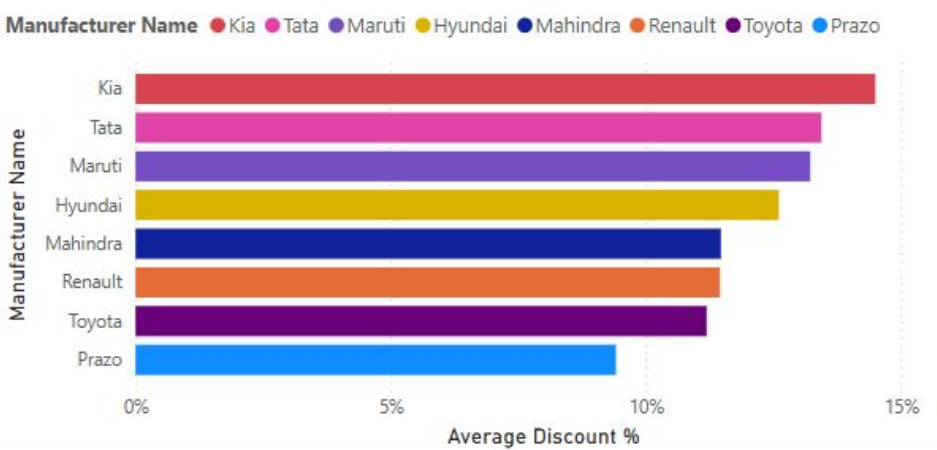
Average Sold Price by Car Type



Average Sold Price by Manufacturer



Average Discount % by Manufactruer



Key Insights

- **Newer cars** sell more frequently, showing strong buyer preference for recent models.
- **Convertibles** have the highest average sold price, while other car types are similar in value.
- **Praz** commands the **highest price** with **lowest discount**, likely a premium brand.
- **Kia** has the **lowest price** and **highest discount**, reflecting a budget-focused strategy.

Conclusion

- The analysis revealed clear trends in used car sales: newer cars sell more often, and manufacturers like Kia tend to offer higher discounts.
- Power BI visualizations helped identify relationships between car type, sale price, and manufacturer performance.
- This project demonstrated the value of combining Excel and Power BI for cleaning, modeling, and drawing insights from sales data.
- These insights can support better pricing, inventory, and marketing decisions in the used car market.